

Key Points

Upon first hearing the idea of Probabilistic Numerics – that computation may be phrased as inference – the critical reader may have some fundamental questions.

- ⊖ Is Probabilistic Numerics even **feasible**? Nice idea, but how might probabilistic numerical methods be built in practice?
- ⊖ Are probabilistic numerical algorithms **fast**? Classic numerical methods are efficient. Bayesian inference has a reputation for high computational cost. Can probabilistic numerical methods be implemented at low complexity?
- ⊖ Do probabilistic numerical algorithms **work**? In fact, what does “works” even mean here? Which analytic desiderata should probabilistic numerical methods satisfy? Classic numerical analysis, most centrally, aims to show that the point estimates of a certain method converge at a desirable *rate* towards the truth. For probabilistic numerical methods, we will likely want a similar notion. But since probabilistic methods add uncertainty as a first-class citizen, we should also be able to make analytic statements showing their uncertainty to be meaningful in some sense.
- ⊖ Does probabilistic mean **stochastic**? After all, Monte Carlo methods, relying on random numbers, are a classic, well-established field of their own.

This chapter uses the elementary problem of integration to establish basic notions and give first exemplary answers to the above questions. In short, these answers are as follows:

- ⊖ It is actually quite straightforward to derive **feasible** instances of probabilistic numerical algorithms. In particular, a

GP prior with a Gaussian covariance yields a first probabilistic numerical quadrature algorithm, producing good estimates for an integral given a set of evaluation nodes. Further, the placement of evaluation nodes across the integration domain, the *actions* taken by the numerical agent, can be naturally controlled by the probabilistic interpretation, by, for example, selecting information gain as a utility function. In this sense, probabilistic numerical methods really are autonomous learning agents.

- ⊖ We will demonstrate, in two different ways, how probabilistic numerical quadrature algorithms can be **fast**. First, many classical numerical algorithms can be directly derived as probabilistic numerical (we will repeatedly do so throughout this book to show connections). Providing an in-depth example, the trapezoidal rule is interpretable as the maximum a posteriori and mean estimator arising from a Wiener process prior on the integrand. Further, the Bayesian inference associated with the trapezoidal rule can be implemented as a Gaussian filter, at a cost identical to that of its classical equivalent. The fact that the trapezoidal rule is probabilistic numerical and the fact that the trapezoidal rule is fast prove that probabilistic numerical algorithms can be fast. We also discuss other common tractable priors for integration, and draw connections to the broader class of (classic) Gaussian quadrature rules. Such connections mean that, instead of having to “re-invent the wheel”, we can use classic methods as inspiration for probabilistic ones. Second, we will show that at least one *ab-initio* probabilistic numerical quadrature algorithm, *warped sequential active Bayesian integration* (WSABI), can be *faster* than standard alternatives. This speed is not despite, but because of, the computation spent on Bayesian inference. Such computation can be seen not as overhead, but as an investment that ultimately yields dividends in returning good estimates more quickly.
- ⊖ Demonstrating what “**works**” means for Probabilistic Numerics, we show convergence statements for both the point estimate and the uncertainty (error estimate) of the probabilistic trapezoidal rule. The most basic uncertainty estimate is found to provide an upper, *worst-case* error bound; and we show that this error estimate can be adapted in an “empirical Bayesian” fashion, at low computational overhead, to provide better calibration, something closer to an *expected-case* estimate.

- ⊖ Probabilistic Numerics is not just different to **stochastic** numerics: Probabilistic Numerics is *opposed* to stochastic numerics. In the univariate setting, we show empirically that Monte Carlo integration has deficiencies over probabilistic numerical deterministic algorithms. We use this example to construct informal, generic arguments against the use of random numbers in computation.

